

7ai	$V_{\text{r.m.s.}} = \frac{9.0}{\sqrt{2}} = 6.4 \text{ V}$	A1
7aii	$T = \frac{2\pi}{\omega} = \frac{2\pi}{20}$ $= 0.31 \text{ s}$	A1
7b	the r.m.s. voltages are different, so no same power dissipated. (Explanation: the r.m.s. voltage for Fig 7.1 is $\frac{V_o}{\sqrt{2}}$ but for Fig. 7.2 it is V_o)	B1
7ci	output power, $P = V_{\text{r.m.s.}} \times I_{\text{r.m.s.}}$ $= 120 \times 0.64$ $= 76.8 \text{ W}$ efficiency $= (76.8/80) \times 100\%$ $= 0.96 \text{ or } 96\%$	C1 A1
7cii	Any one from:	B1

	• heat losses due to resistance of windings / coils • heat losses in magnetising and demagnetising core / hysteresis losses in core • heat losses due to eddy currents in (iron) core • loss of flux in the (iron) core	
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8ai	By <u>Fleming's Left Hand Rule</u> , the force acting on the wire is <u>out of the plane</u> of paper. By <u>Newton's Third Law</u> , the force on the magnet is <u>into the plane</u> of paper.	B1 B1
8aii	Force decreases	M1